

Module 11: Distance Vector Routing and Count-to-Infinity

Unit 3: Network Layer

Bellman-Ford & Routing Dynamics

Mathematical Foundation of Distance Vector Routing, Bellman-Ford Recurrence, The Count-to-Infinity Failure Phenomenon, and Split Horizon / Poison Reverse Mitigations.

Distance Vector Routing (Bellman-Ford) & Count-to-Infinity Problem

$$\text{Bellman-Ford Equation: } D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$$

Linear 3-Node Topology (RIP Protocol)



Stable Routing Vectors (to Destination A):

- At Node B: Distance to A = 1 (Direct via link B-A)
- At Node C: Distance to A = 2 (Next Hop = B)

Sudden Link Failure: Link A-B Goes DOWN!

- B detects link A-B is dead.
- But before B can advertise this, C tells B: "I have route to A (cost=2)!"
- B mistakenly thinks: "Great! I can reach A via C with cost = 2+1 = 3!"
- Routing Loop created: B points to C, C points to B!

Count-to-Infinity Ping-Pong Cycle

Exchange Step	B's Cost to A	C's Cost to A
Iteration 1	3 (via C)	2 (via B)
Iteration 2	3 (via C)	4 (via B)
Iteration 3	5 (via C)	4 (via B)
...	Continues until metric reaches 16 (Infinity)	

Standard Engineering Solutions:

- Defining Infinity (RIP = 16):**
Caps maximum loop counting at 16 hops.
- Split Horizon:**
Never advertise a route back to the neighbor from whom it was learned!
- Split Horizon with Poison Reverse:**
Explicitly advertise the reverse route with cost = infinity (16).

1. Bellman-Ford Recurrence Relation

$$D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$$

Periodic Updates: Broadcast entire routing table to direct neighbors every 30 seconds.

Metric Cap: RIP defines Infinity = 16 hops to limit count-to-infinity duration.

2. Count-to-Infinity Mitigations

Technique	Core Mechanism	Tradeoff
Hop Count Cap (16)	Caps maximum routing loop iterations	Limits network diameter to 15 hops
Split Horizon	Never advertise a route back to the advertising neighbor	Prevents 2-node loops
Poison Reverse	Explicitly advertises reverse route with metric = Infinity (16)	Instant loop elimination